

liquidity.⁹⁹⁴ Both price competition and adverse selection from pennyng lie on a continuum.⁹⁹⁵ As explained in *infra* section VII.D.1, the degree to which pennyng versus price competition dominates in determining whether increasing the tick will improve market quality depends on the relation between the tick and the spread. The greater the tick is in relation to the spread, the greater the effect of price competition, and the lower the risk of pennyng.⁹⁹⁶ Accordingly, the Commission defines a stock to be tick-constrained if there is a reasonable probability that the stock would otherwise trade with a spread less than the tick size in the course of normal trading, were it allowed to do so, or one for which the tick is a substantial portion of the quoted spread.⁹⁹⁷

⁹⁹⁴ *Id.* at 80305-06. Pennyng is defined in the Proposing Release as occurring when a market participant gets to the front of the limit order queue by posting economically trivial price improvement. *Id.* at 80306 n.459. One commenter also described the economics of pennyng using option theory and the Commission agrees with this characterization. *See* Harris Letter at 6. According to the commenter, pennyng results in a payoff structure that has unlimited potential upside while the downside is capped. By pennyng, a fast trader jumps to the front of the queue and therefore has a high chance of executing his trade and capturing the upside if prices move favorably. If prices move in an unfavorable direction, the fast trader can unwind his position against the slower liquidity supplier (whom he undercut); in this case, the cost to the fast trader—*i.e.*, the cost of the option—is only one tick. The fast trader thereby captures value from the liquidity supplier and hence discourages slow traders from offering liquidity. A small tick means that the cost of pennyng is low, which results in more pennyng and thus less incentive for liquidity provision. *See also* Lawrence E. Harris, *Minimum Price Variations, Discrete Bid-Ask Spreads, and Quotation Sizes*, 7 REV. FIN. STUD. 149 (1994); Anne Dyhrberg, et al., *When Bigger is Better: The Impact of a Tiny Tick Size on Undercutting Behavior*, 58 J. FIN. & QUANTITATIVE ANALYSIS (2023) (Dyhrberg et al.).

⁹⁹⁵ Proposing Release, *supra* note 11, at 80305 n.458.

⁹⁹⁶ *See infra* section VII.D.1.b.i.

⁹⁹⁷ The Commission’s definition of tick-constrained in this release eliminates an unnecessary distinction drawn between tick-constrained and near-tick-constrained stocks that appears in the Proposing Release. Specifically, in the Proposing Release’s economic analysis, the Commission stated that it considered the term “tick-constrained” to apply to “stocks that would otherwise trade with a spread less than the tick size, were they allowed to do so.” Proposing Release, *supra* note 11, at 80304. That economic analysis also stated that a “near-tick-constrained” stock was “one that has a reasonable probability of becoming tick-constrained in the course of normal trading, or one for which the tick is a substantial portion of the spread.” *Id.* Given the economics of minimum pricing increments discussed in this section, distinguishing near-tick-constrained stocks from tick-constrained stocks is unnecessary. The Proposing Release’s discussion of these terms in its economic analysis did not meaningfully differentiate between the effects on stocks in each of these groups. As a result, applying the singular term tick-constrained avoids confusion and streamlines the discussion. In addition, the Proposing Release’s empirical analysis employed separate numerical definitions for tick-constrained and near-tick-constrained stocks. *Compare id.* at 80268 n.17 (tick-constrained stocks are those with a Time Weighted Average Quoted Spread less than .011) with *id.* at 80304 n.449 (near-tick-constrained stocks are those with a Time Weighted Average Quoted Spread

That is, while tick constrained stocks are not the only ones to potentially benefit from a reduction in the tick size, they are the ones most clearly likely to do so.⁹⁹⁸

Benefits from a reduction in the tick size come in the form of higher market quality and lower transaction costs to investors. As explained above, relaxing the tick constraint (the price floor on liquidity) directly allows competition for market orders. Moving toward a more competitive market reduces distortions and economic rents.

As a general matter, a liquidity provider is incentivized to get its quote to the front of the queue (*i.e.* establish price/time priority on an order book).⁹⁹⁹ This is because stock exchange priority rules give greater priority to better priced orders and generally factor order entry time into the priority of limit orders at the same price. When the NBBO is equal to the tick, liquidity providers cannot establish price priority (other than by crossing the spread)¹⁰⁰⁰ because there are no price points at which to do so.¹⁰⁰¹ Because liquidity providers cannot establish price priority

between .011 and .02). These numerical definitions served as proxies for drawing distinctions in the Proposing Releases' empirical analysis. See, e.g., id. at 80304 nn.448-449; at 80308 n.473 and accompanying text; and at 80319 n.549. Although we continue to include specific explanations of what stocks are included in each of our quantitative analyses where relevant, to further simplify the discussion in the economic analysis, we do not use the definitions employed in the empirical analysis more broadly.

⁹⁹⁸ See Proposing Release, supra note 11, at 80309 and the discussion accompanying nn.474-477. See also CCMR Letter at 18 ("an MPI that is too wide may set an artificial constraint on permissible bids and offers, which can result in an unnecessarily wide spread that can also increase transaction costs for investors").

⁹⁹⁹ Material presented in this paragraph was discussed in the Proposing Release, supra note 11, at 80309 and nn.478-481.

¹⁰⁰⁰ "Crossing the spread" refers to switching from posting a (non-marketable) limit order to sending a market order. For example, if the national best bid were \$10.00 and the national best offer were \$10.02, a limit order to buy, if executed, would entail paying \$10.00 for the security. However, a market order to buy would entail pay \$10.02, in other words, it would have crossed the spread of \$0.02. In the most common case of maker-taker, the difference between the market and limit order is even greater because the buyer using a market order would pay \$10.02 plus the fee, whereas the buyer using a limit order would pay \$10.00 minus the rebate.

¹⁰⁰¹ The liquidity provider could submit an order at an inverted exchange, though this is an inefficient solution. See section VII.C.2.c.

when the NBBO spread is one tick, establishing time priority becomes more important.¹⁰⁰²

Consequently, an environment where stocks are tick-constrained with artificially wider spreads and longer order queues tends to favor traders who are better able to establish positions more quickly so they can be at the front of the queue. Traders who are at the back of the queue face slower executions and the risk of not being executed against at all (a lower fill rate). In the latter case, they will need to resubmit an order when the market has moved in an unfavorable direction, increasing transaction costs. Adverse selection amplifies these costs: the orders of slower traders are most likely to be executed when such an execution is unfavorable to them and least likely when they would be favorable. For example, a sell order at the back of the queue will tend to be filled when there are many buy orders, which tend to increase the price, implying that selling is disadvantageous.

To summarize, current wider quoted spreads mean greater cost to liquidity demanders and greater revenue to liquidity providers.¹⁰⁰³ An artificially wide spread, due to a price floor imposed by the tick constraint, effectively subsidizes liquidity provision. Because there is an

¹⁰⁰² Under typical exchange rules, an order with time priority is executed first when multiple orders are at the best price, regardless of how many orders are at the best price. In longer order queues, liquidity-providing orders deeper in the queue, which do not have time priority, are less likely to be filled in a timely manner and, conditional on being filled, the probability of the order having been adversely selected tends to be greater compared to orders with greater fill priority. Typically, liquidity providers compete to gain priority over other resting orders by quoting a better price, but tick-constraints make doing so difficult. In the case when the spread is constrained to a single tick, it would be impossible to improve on the displayed price without locking markets. For tick-constrained stocks, when the quoted spread may be greater than a single tick, improving the price by an entire tick may be too much in the sense that doing so may narrow the spread beyond what the liquidity providers could tolerate. A narrower tick de-emphasizes time priority on a stock exchange by making it easier to compete on price. See Edwin Hu, et al., Tick Size Pilot and Market Quality (DERA White Paper, Jan. 31, 2018), available at https://www.sec.gov/dera/staff-papers/white-papers/dera_wp_tick_size-market_quality; and Todd G. Griffith & Brian S. Roseman, Making Cents of Tick Sizes: The Effect of the 2016 U.S. SEC Tick Size Pilot on Limit Order Book Liquidity, 101 J. BANKING FIN. 104 (2019).

¹⁰⁰³ Market participants can use inverted exchanges or ISOs to help ameliorate some of the negative effects of tick size constraints.

increased incentive to provide liquidity via limit orders, queues of limit orders tend to be longer, and wait times to get a limit order executed also tend to be longer. This makes it more likely that the market moves away from an investor's limit order and leads to lower overall fill rates for limit orders.¹⁰⁰⁴ Thus the floor on liquidity leads to rents accruing to fast liquidity providers¹⁰⁰⁵ at the expense of slower ones as well as liquidity demanders.¹⁰⁰⁶

3. Economics of Access Fees

Trading venues can choose to charge an access fee, or pay a rebate, to their participants—liquidity providers and liquidity takers—who trade at their venue. The trading venue can further choose to levy the fee (or pay the rebate) on either the liquidity taker or liquidity provider, or on both. As discussed in *infra* section VII.C.2.b, the most common fee structure is maker-taker, in which liquidity takers are assessed an access fee and liquidity providers are paid a rebate, which is typically funded through the access fee. That is, for a buy order the liquidity taker pays the

¹⁰⁰⁴ See, e.g., Barbara Rindi & Ingrid M. Werner, U.S. Tick Size Pilot (working paper Mar. 4, 2019), available at <https://ssrn.com/abstract=3041644> (retrieved from SSRN Elsevier Database); Mao Ye & Chen Yao, Tick Size Constraints, Market Structure and Liquidity (working paper Dec. 26, 2019), available at <https://ssrn.com/abstract=2359000> (retrieved from SSRN Elsevier database); Phil Mackintosh, Why Ticks Matter, NASDAQ (May 19, 2022), available at <https://www.nasdaq.com/articles/why-ticks-matter>; and MEMX, Tick-Constrained Securities (Aug. 2021) (“MEMX Report”), available at <https://memx.com/wp-content/uploads/MEMX-Market-Structure-Report-Tick-Constrained-Securities.pdf>.

¹⁰⁰⁵ This phenomenon is sometimes referred to as excessive intermediation. In this context, excessive intermediation refers to excessive quoting in sufficiently liquid securities in order to profit from the tick-constraint-induced price floor on liquidity, which crowds out investors from being able to supply liquidity. Such price floors can increase quoting activity from high-frequency traders looking to earn the artificially high spread. Because profiting off of the spread is easiest when the marketable orders filled are small, obtaining high priority in the queue at each tick is essential to such strategies. High-frequency, proprietary traders are generally better able to obtain such priority, and consequently investors may have less opportunity to profitably fill their trades using limit orders. Rebates on limit orders further increase the incentives of these traders to engage in such intermediation, thereby exacerbating the problem.

¹⁰⁰⁶ In support of this point, one commenter stated that the subsidization of liquidity providers resulting from the tick constraint leads to greater competition on the basis of speed to provide liquidity, which increases complexity and related costs to investors; See Budish Letter at 4.

price plus the access fee. For a sell order, the liquidity taker receives the price, less the fee.

Assuming the broker-dealer is the principal to the trade, then the economic price of accessing or providing liquidity would be equivalent to the displayed nominal price net of the applicable fees.

If the broker-dealer is an agent, then there may a wedge between economic price of accessing or providing liquidity and the price net of the fee and rebate. It is possible that the fee and rebate may be passed on directly to the customer. The fee and rebate may be passed on indirectly and in part through fees, commissions, or as part of a bundle of services to the customer.

Section VII.C.2 describes the current market structure as it relates to access fees and rebates. A key feature of the current market structure is that many exchanges charge at the current cap and pay out nearly all of the fee as a rebate. For this reason, in practice the Commission expects access fees to be near the cap under the amended rule, just as they are near the pre-existing cap under the current structure. As discussed in the Proposing release, several basic economic considerations are among those governing the analysis of access fees. First, access fees should be such that net and quoted prices satisfy coherence.¹⁰⁰⁷ Second, under simplifying assumptions, fees and rebates are approximately neutral provided that the stock is not tick constrained,¹⁰⁰⁸ although outside of those simplifying assumptions lowering the access fee cap can have additional benefits as discussed in section VII.D.2.d. Finally, for tick constrained stocks, access fees and rebates can distort liquidity supply and demand, increasing transaction costs for investors.¹⁰⁰⁹

¹⁰⁰⁷ See Proposing Release, supra note 11, at 80348 (“Net and nominal price rankings are coherent if sorting trading venues on the competitiveness of their nominal quoted prices yields the same ordering as sorting on prices net of fees and rebates.”).

¹⁰⁰⁸ Proposing Release, supra note 11, at 80328.

¹⁰⁰⁹ Id.

In response to the Proposing Release, the Commission received extensive comment regarding the role of fees and rebates on the supply of and demand for liquidity.¹⁰¹⁰ Below, to address comments, the Commission supplements its discussion on how access fees and fee-funded rebates may affect trading in the presence of the commonly used maker-taker fee structure.¹⁰¹¹ This section addresses certain aspects of fees and rebates in developing a basic framework for evaluating the principle economic effects and responding to comments. The remaining aspects and effects are considered in sections VII.C.2, VII.D.2, and VII.D.3.

a. Liquidity with Access Fees and Rebates

In the absence of ticks, the market for liquidity, discussed in section VII.B.1, may generally be represented by an economic model of supply and demand, as shown below in Panel A of Figure 1.¹⁰¹² The vertical axis represents the price of liquidity, here the quoted half-spread

¹⁰¹⁰ Specifically, commenters had different views on whether reducing access fees and rebates will adversely affect the provision of liquidity on exchanges, either generally or for particular categories of stocks. Compare Cboe Letter II at 9; Cboe Letter III at 6, 8; Cboe Letter IV at 3, 5; CCMR Letter at 27; IEX Letter I at 2; IEX Letter IV at 18, 21; IEX Letter V at 4; IEX Letter VI at 7; Nasdaq Letter I at 2, 20, 22, 25; Nasdaq Letter II at 3, 6; Nasdaq Letter III at 2-3; Themis Letter at 7-8; Virtu Letter II at 7-8. Commenters likewise had differing perspectives on whether reducing access fees and rebates will reduce overall transaction costs, thereby increasing demand for liquidity, or cause offsetting costs related to, e.g., wider spreads, volatility, or a less representative NBBO (which could reduce demand for liquidity). Compare Cboe Letter III at 5-6; Cboe Letter IV at 5; CCMR Letter at 27; IEX Letter I at 26; IEX Letter IV at 16, 22-23; Nasdaq Letter I at 2, 20, 22-24; Nasdaq Letter II at 4-7; Nasdaq Letter III at 5; Themis Letter at 7; Virtu Letter II at 8. As one example of this debate, Nasdaq identifies a “vicious cycle” that could result from a reduction in the minimum access fee, whereas IEX identifies a “virtuous cycle” from the identical change. See Nasdaq Letter II at 6; IEX Letter I at 26. See section VII.D.2 for a response to these comments on the effect of the reduction in the access fee cap on liquidity and transaction costs.

¹⁰¹¹ The economic theory laid out below allows the Commission to create a common framework for the competing claims of commenters and to disentangle the complex forces at work in determining spreads. While the Commission has supplemented this discussion from the Proposing Release, the essential point in this framework—the equilibrium resulting from the supply and demand for liquidity, modified as necessary for the presence of a minimum tick—was discussed throughout the Proposing Release. See, e.g., Proposing Release, supra note 11, at 80228-29, 80317, 80321, 80336, 80338.

¹⁰¹² This model presents an abstraction of the market for liquidity. As explained in section VII.B.2, the willingness to quote a bid or offer depends in part on the degree of adverse selection in the market which in turn depends on the tick size. The key point in this section is that, for stocks that have a spread that is sufficiently wide, liquidity providers are indifferent between receiving compensation in the form of spread

(*i.e.* because the liquidity taker is not typically on both sides of the trade, we use the quoted half-spread to measure price of liquidity¹⁰¹³), while the horizontal axis represents the quantity of liquidity.¹⁰¹⁴ Liquidity providers supply liquidity, and the supply curve is upward sloping because liquidity providers are willing to supply more liquidity when the price of liquidity is higher. Liquidity takers demand liquidity, and the demand curve is downward sloping because liquidity takers demand less liquidity at higher prices of liquidity (*i.e.*, they trade less when they have to pay higher transaction costs). The supply and demand curves intersect at the point where the amount of liquidity supplied equals the quantity demanded, which indicates the equilibrium price and quantity of liquidity in the market.

or in the form of a rebate; similarly, liquidity demanders are indifferent between paying the spread or access fee, and thus fees and rebates tend to be neutral assuming a spread that is sufficiently wide. This point is unaffected by the presence of adverse selection arising from a tick that may be too narrow.

¹⁰¹³ The price of liquidity is represented by the quoted half-spread because the half-spread represents the price which liquidity takers must pay for the immediacy of executing their trade while liquidity providers stand to capture the half-spread.

¹⁰¹⁴ Figure 1 shows the supply and demand for liquidity, with the quoted half-spread as representing the price of liquidity on the y-axis. This should not be confused with supply and demand for shares of the stock, where the price of the stock would be on the y-axis.

Figure 1

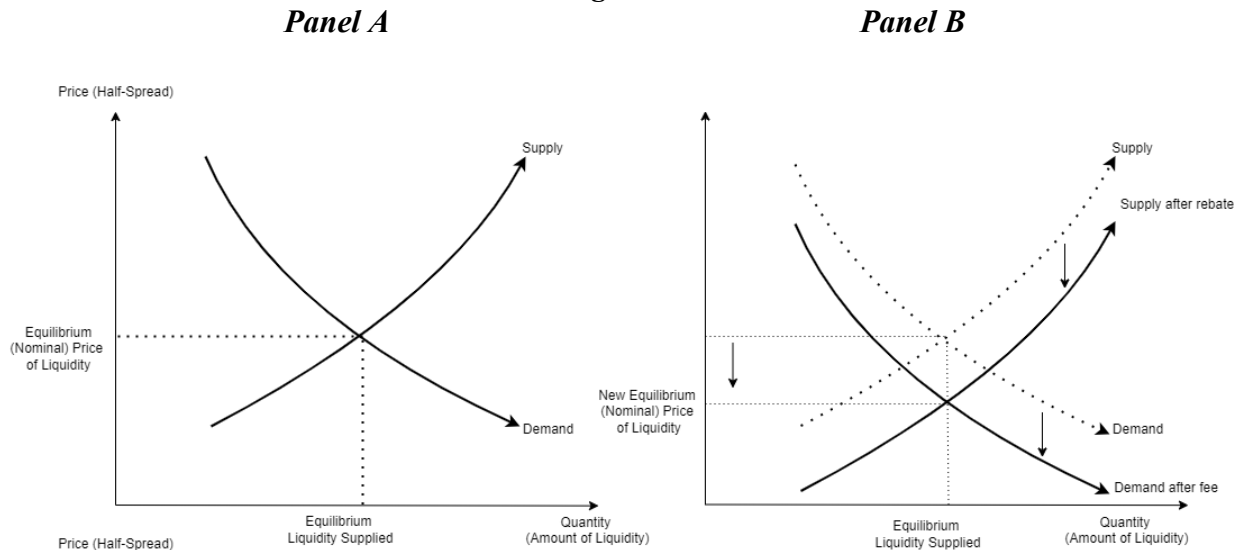


Figure 1: The above figures illustrate the market for liquidity represented by a supply curve and a demand curve. Liquidity suppliers contribute to the supply of liquidity by posting passive orders such as limit orders, while liquidity demanders trade against the passive orders thereby taking liquidity. All other things equal, the quantity of liquidity supplied increases as the price of liquidity increases, while the quantity of liquidity demanded decreases as the price of liquidity increases. The price of liquidity is represented by the quoted half-spread because the half-spread represents the price which liquidity takers must pay for the immediacy of executing their trade while liquidity providers stand to capture the half-spread. The intersection of these curves sets the equilibrium price of liquidity (i.e., the half-spread) and the equilibrium quantity of liquidity. As demonstrated in the supply-demand diagram on the right, rebates shift the supply curve so that there is more supply for a given quoted price of liquidity. Fees increase the cost of demanding liquidity so the shift in the demand curve reflects that there will be less demand at any given price of liquidity. Because both supply and demand curves vertical shifts are parallel, they continue to intersect at the same value along the horizontal axis (the amount of liquidity supplied/demanded).

Consider next the effect on liquidity of a 30 mils access fee used to fund a 30 mils rebate.¹⁰¹⁵ With a rebate of 30 mils, liquidity providers who submit buy orders are willing to increase their bid price by 30 mils, while liquidity providers who submit sell orders are willing to lower their ask price by 30 mils. For this reason, the bid-ask spread narrows by 60 mils, and the quoted half-spread (*i.e.*, price of liquidity) narrows by 30 mils. Accordingly, in Panel B of Figure

¹⁰¹⁵ This level of access fee and rebate is similar to current fees and rebates on maker-taker lit exchanges and may vary based on pricing tier; see *infra* section VII.C.2.b. To illustrate the salient economic points, this discussion assumes that liquidity providers and demanders know what the resulting access fees and rebates from a transaction will be. As discussed below in sections VII.C.2.b and VII.D.3, in the baseline this is only ever approximately true since fees and rebates are often determined using current and future volumes. But as long as market participants are able to approximate their fees and rebates, then they will generally behave as described in this paragraph—liquidity providers will adjust their quotes on account of the expected rebate, and liquidity demanders will adjust the quoted price they are willing to pay on account of the expected fee.

1, the supply curve for liquidity shifts down by 30 mils. Likewise, the 30 mils access fee acts as a tax on liquidity takers. This means that, for the same amount of liquidity, liquidity takers will reduce the price they are willing to pay by 30 mils to account for the access fee (since their net cost to take liquidity is the price they pay plus the access fee). In Panel B of Figure 1, this effect is represented by the liquidity demand curve shifting down by 30 mils. Because both the supply curve and the demand curve shift down by 30 mils, they continue to intersect at the same quantity of liquidity.¹⁰¹⁶ That is, the equilibrium amount of liquidity remains unchanged, but the displayed price is 30 mils lower. The net cost to take liquidity is not affected since it equals the price of liquidity plus the 30 mils access fee;¹⁰¹⁷ similarly, the net proceeds from providing liquidity are not affected since they equal the price of liquidity plus the 30 mils rebate.¹⁰¹⁸ Thus, in the absence of frictions (*e.g.*, discrete prices, minimum pricing increments, or agency problems), the level of fees and rebates (when fees and rebates are equal in size) does not affect the total costs of trading.¹⁰¹⁹ As one commenter put it, “when liquidity suppliers are subsidized at the cost of liquidity takers, spreads decline. If they did not, everyone would want to be a liquidity

¹⁰¹⁶ If the demand curve were vertical (namely if liquidity demanders were not sensitive to price), the curve would not shift. However, the conclusions would be the same in that the supply curve shift would cause the same quantity to be supplied at a lower price.

¹⁰¹⁷ The concept that net cost (or net spread) is the correct way to measure the cost of liquidity is supported by basic economics and by commenter statements. See Citigroup Letter at 6 (stating, “Many of CGMI’s institutional clients are increasingly measuring their execution costs all-in, inclusive of exchange fees.”).

¹⁰¹⁸ This discussion refers to the cost to take liquidity and proceeds from providing liquidity at the time of the execution of the trade.

¹⁰¹⁹ The term frictions here refers to factors that prevent prices from perfectly reflecting the forces of liquidity supply and demand. It does not imply that a frictionless market is the optimal market construct. As discussed throughout this release, a tick size that is too small creates pennyng concerns which can harm market quality outcomes. See *infra* section VII.D.1.b for additional discussion.

supplier, and no trade would occur. So, maker-taker pricing created narrower quoted spreads on average, but it does not affect the net cost of providing liquidity.”¹⁰²⁰

Academic work on the effect of a fee change on the Toronto Stock Exchange supports the model.¹⁰²¹ In 2005, the exchange began offering a rebate to liquidity suppliers in a pre-defined subset of securities. For securities in which the total fee remained constant but was split into a maker rebate and a taker fee, the authors find that quoted spreads narrow, but the net spread—which includes the quoted spread and the take fee—did not change.¹⁰²²

b. Liquidity with Ticks, Access Fees, and Rebates

Section VII.B.3.a shows that the quoted spread adjusts to a fee and rebate by narrowing by the amount of the fee and the rebate. The supply and demand curves in section VII.B.3.a are continuous, whereas in fact displayed liquidity has discrete price points, namely ticks (the focus of section VII.B.2). In the presence of ticks, access fees and rebates need no longer be neutral, namely the quoted spread may not adjust in the same seamless way to the presence of a rebate. The basic intuition of section VII.B.3.a states that a liquidity provider is indifferent between receiving compensation from the spread and compensation from the rebate. If a rebate is offered, the liquidity provider in a competitive market responds by accepting a lower spread. However, if the quoted spread is already at its floor, as specified by the tick, it is not possible to further lower

¹⁰²⁰ Harris Letter at 2-3, describing the equilibrium spreads model (citing Kalman J. Cohen, et al., Transaction Costs, Order Placement Strategy and Existence of the Bid-Ask Spread, 89 J. POL. ECON. 287 (1981)).

¹⁰²¹ See Katya Malinova & Andreas Park, Subsidizing Liquidity: The Impact of Make/Take Fees on Market Quality, 70 J. FIN. 509 (2015).

¹⁰²² The authors also present evidence suggesting that adverse selection costs decreased with the introduction of the maker-taker model by increasing retail trader participation. See Malinova and Park (2015), *supra* note 1021. In the United States, most retail orders in NMS stocks are handled by wholesalers, who execute a large majority of the dollar volume of the retail orders they handle via internalization. See, e.g., Lewis Letter attached to Virtu Letter II at p 8-12 and 40-45. Consequently, should there be a decrease in retail participation, we do not expect this to cause an increase in adverse selection on exchanges, because so much of retail order flow passes through a wholesaler before being executed.